ASSIGNMENT 3

3. If μ= 55, σ_{4a}= 4, $\sigma_{4\beta}$=10, σ_4=c 15, In this which is better

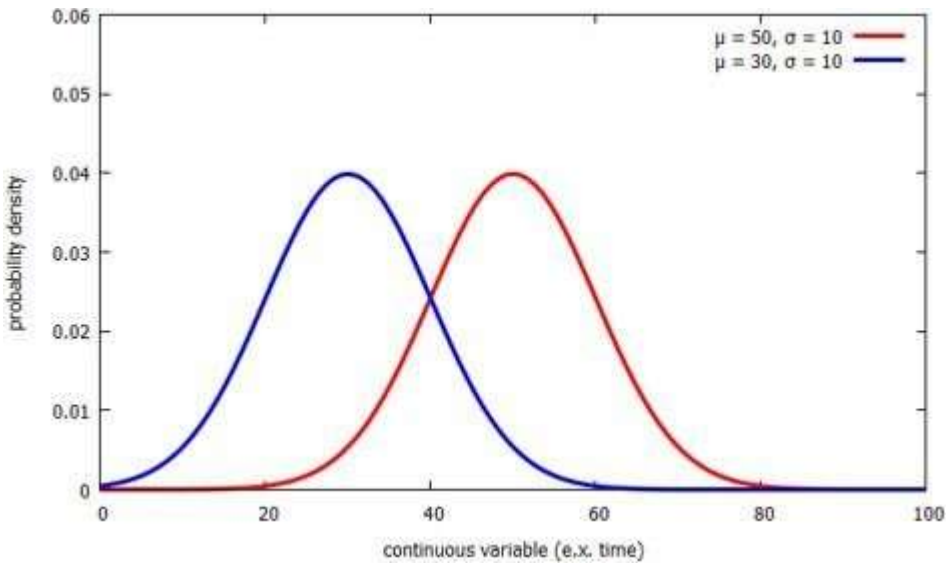
Comparison of Normal Distributions with Same Mean and Different Standard Deviations

(μ = 55, σ_a = 4, σ_β = 10, σc = 15)

1. Introduction

In statistics and data analysis, the behavior of a dataset is mainly described using two parameters: **mean (μ)** and **standard deviation (σ)**. The mean represents the central value around which data is distributed, while the standard deviation measures the spread or variability of data around that mean. When multiple datasets have the same mean but different standard deviations, comparison of their spreads helps in identifying which dataset is more stable, reliable, and desirable.

In this case, three normal distributions have the same mean μ = 55 but different standard deviations: σ_a = 4, σ_β = 10, and σc = 15. This theory-based discussion explains which distribution is better and why.



- Statistics is the science of collecting, analyzing, interpreting, and presenting data.
- In statistical analysis, understanding how data behaves is crucial for decision-making.
- Two fundamental measures that describe any dataset are **mean** and **standard deviation**.
- A **normal distribution** is one of the most important probability distributions

used in statistics.

2. Meaning of Normal Distribution

- A normal distribution is a continuous probability distribution.
- It has a bell-shaped curve.
- The curve is symmetric about the mean.
- The left and right halves of the curve are mirror images.
- The total area under the curve equals 1.
- It is completely defined by two parameters:
  - Mean (μ)
  - Standard Deviation (σ)

3. Role of Standard Deviation

Standard deviation indicates how much individual data values differ from the mean. It directly affects the shape of the normal distribution curve:

- A **small standard deviation** results in a **narrow and tall curve**, showing that data points are closely clustered around the mean.
- A **large standard deviation** produces a **wide and flat curve**, indicating greater dispersion and variability in the data.Thus, standard deviation is a key factor in assessing consistency and predictability.

5. Case A: $\sigma_a = 4$

- Represents very low variability.
- Data values are tightly clustered around the mean.
- The normal curve is tall and narrow.
- Indicates high consistency and high reliability.
- Best suited for stable systems.

6. Case B: $\sigma_\beta = 10$

- Represents moderate variability.
- Data values are moderately spread around the mean.
- The normal curve is neither too narrow nor too wide.
- Indicates average consistency and moderate reliability.

7. Case C: σ_C = 15

- Represents high variability.
- Data values are widely spread.
- The normal curve is flat and wide.
- Indicates low consistency and low predictability.
- Extreme values are more likely to occur.

8. Graphical Interpretation

- All three curves are centered at μ = 55.
- σ_a = 4 → Tall and narrow curve.
- σ_β = 10 → Medium-width curve.
- σ_C = 15 → Wide and flat curve.
- As σ increases, the spread of data increases.

G.MEAN AND ITS LIMITATIONS

What is Mean?

Mean is the central value of a dataset.

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Limitation of Mean:

- Does not show variability
- Cannot indicate consistency
- Cannot show reliability alone

◆ **Dataset A (σ_{4_a} = 4)**

$$CV_A = \frac{4}{55} \times 100 = 7.27\%$$

✓ Very low variation
✓ Highly consistent

◆ **Dataset B ($\sigma_{4\beta}$ = 10)**

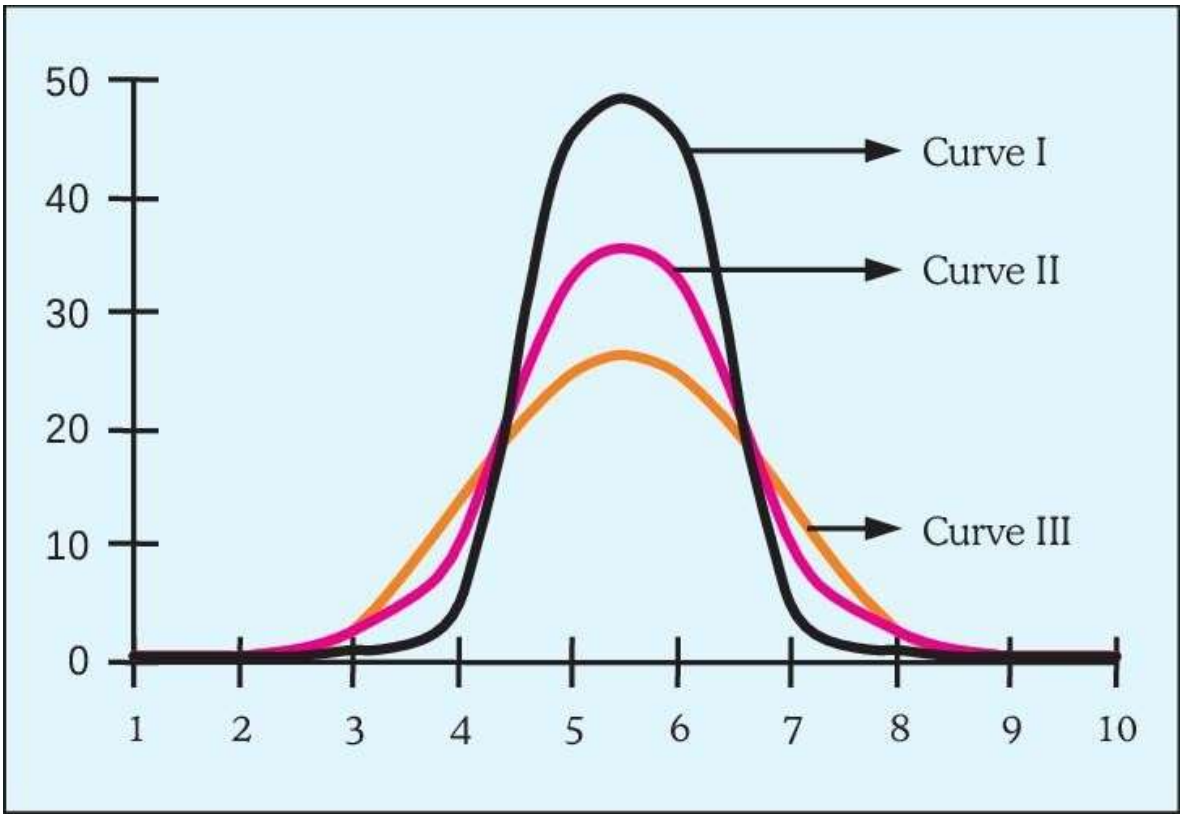
$$CV_B = \frac{10}{55} \times 100 = 18.18\%$$

✓ Moderate variation
✓ Less consistency than A

◆ **Dataset C (σ_{4C} = 15)**

$$CV_C = \frac{15}{55} \times 100 = 27.27\%$$

+ High variation
+ Least reliable



Given Data (Restated Clearly)

- Mean (μ) = 55
- Distribution A: $\sigma_a = 4$
- Distribution B: $\sigma_\beta = 10$
- Distribution C: $\sigma c = 15$

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8. Case A: Distribution with $\sigma_a = 4$

- Very small standard deviation.
- Data points are tightly clustered around the mean.
- Curve is tall and narrow.
- Most values lie close to 55.
- Very few extreme values.
- Indicates strong consistency.
- Indicates high stability.
- Indicates high reliability.
- Suitable for systems requiring precision.

G. Advantages of Case A ($\sigma = 4$)

- Low variability.
- High predictability.
- Better control over outcomes.
- Less uncertainty.
- Better performance consistency.
- Ideal for quality-sensitive environments.

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10. Case B: Distribution with $\sigma_\beta = 10$

- Moderate standard deviation.
- Data points are moderately spread.

- Curve is neither too narrow nor too wide.
- Some values deviate noticeably from the mean.
- Indicates average consistency.
- Indicates moderate reliability.
- Suitable for systems where some variation is acceptable.

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11. **Characteristics of Case B**

- Balanced distribution.
- Neither highly stable nor highly unstable.
- Moderate level of risk.
- Performance is acceptable but not optimal.

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12. **Case C: Distribution with $\sigma_C = 15$**

- Very large standard deviation.
- Data points are widely spread.
- Curve is flat and wide.
- Many values lie far from the mean.
- Extreme values are common.
- Indicates poor consistency.
- Indicates low reliability.
- High uncertainty in outcomes.

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13. **Disadvantages of Case C**

- Unpredictable results.
- Large deviations from average.
- Higher risk.
- Poor control over data behavior.
- Not suitable for precision-based systems.

14. **Graphical Interpretation**

- All three curves are centered at $\mu = 55$.
- Case A curve is tallest and narrowest.
- Case B curve is moderately wide.
- Case C curve is widest and flattest.
- Increase in σ spreads the curve horizontally.
- Height of the curve decreases as σ increases.

15. **Comparison Based on Consistency**

- Case A: Very high consistency.
- Case B: Moderate consistency.
- Case C: Low consistency.

16. **Comparison Based on Reliability**

- Case A: Most reliable.
- Case B: Moderately reliable.
- Case C: Least reliable.

17. **Comparison Based on Risk**

- Case A: Minimal risk.
- Case B: Moderate risk.
- Case C: High risk.

18. **Conclusion**

- When the mean remains constant, the distribution with the smallest standard deviation is the most desirable. In this case, the normal distribution with $\mu = 55$ and $\sigma_a = 4$ is better than those with $\sigma_\beta = 10$ and $\sigma c = 15$ because it shows the least variability and highest consistency. Therefore, **$\sigma_a = 4$ represents the best distribution among the given options**.